DSO 576 | Module 6 Project Data Cleaning

Project: Emerging Artist Discovery Student: Kunjesh Gupta

Vision and question

Editorial curation lead at Spotify must decide which emerging artists to add to discovery playlists and which listener segment to surface each one to.

Cleaning question: Can a cleaned ListenBrainz event sample support descriptive comparisons of artist listen counts and unique listeners by month? The sample is not representative of Spotify users and does not support causal claims.

Source and reproducibility

Public source: https://api.listenbrainz.org/1/user/hwnrwx/listens?count=50 (user hwnrwx, requested count 50). Raw file SHA-256: 04507f491c3a7c2cc3321da8b0476e23227345058836d8d39fbc05ab19708ac6. Run: `python3 cleaning_kunjesh/cleaning.py --input cleaning_kunjesh/listenbrainz_sample.json`. Expected outputs are in `cleaning_kunjesh/output/`.

14-item checklist

#	Item	Status	Evidence
1	State project vision and question	Done	This PDF, Vision and question
2	Define one row	Done	cleaning.py flatten_listens; source_index
3	Preserve source	Done	Raw JSON and SHA-256 above
4	Inspect columns and types	Done	cleaned_listens.csv and conversion flags
5	Standardize text	Done	*_raw plus cleaned text columns
6	Investigate duplicates	Done	0 exact duplicates removed; repeated IDs retained
7	Handle missing data	Done	missingness.csv and metric exclusion flag
8	Check suspicious values	Done	invalid/future timestamp and MSID flags
9	Verify merges	Not applicable	No merge in event-level pass
10	Reconcile changes	Done	50 starting -> 50 final rows
11	Check calculations	Done	metrics.csv includes listen and unique-listener denominators
12	Verify individual records	Done	Five checks on next page
13	Test reproducibility	Done	Run command above; generated outputs
14	Document limitations	Done	Limitations below

Cleaning decision log

Rule	Decision	Check	Status
one row	Each row is one ListenBrainz listen event; source_index identifies its original position.	Payload structure and API JSON documentation	Done
preserve source	Keep the downloaded JSON unchanged and write derived CSV outputs separately.	raw JSON and SHA-256 recorded in README	Done
text	Trim and collapse whitespace; retain original values in *_raw columns.	raw and cleaned text columns	Done
timestamps	Coerce listened_at to UTC; exclude missing timestamps from time metrics.	invalid_timestamp and excluded_from_metrics	Done
duplicates	Remove only exact payload duplicates; retain repeated IDs because repeated listens are valid events.	exact_duplicate and row-count reconciliation	Done
identifiers	Preserve MBIDs/MSIDs as strings and flag malformed recording MSIDs.	invalid_recording_msid	Done
merges	No merge is performed in this event-level cleaning pass.	No join keys or extra tables used	Not applicable
Codex check	Codex suggested excluding missing-time events from time metrics; verify with excluded-row counts and metric denominators.	excluded_from_metrics and metrics.csv	Done

Missingness before and after cleaning

Column	Before	After
user_name	0	0
listened_at	0	0
artist_name	0	0
track_name	0	0
recording_msid	0	0
artist_mbid	50	50
recording_mbid	50	50
release_mbid	50	50

Five record checks

Expected outcomes were defined from the cleaning rules before checking the generated output.

Source index	Original	Expected	Actual	Match
0	hwnrwx | 1600152461 | Legião Urbana	UTC timestamp parses; user and artist text are trimmed; row remains a valid listen event.	timestamp=2020-09-15 06:47:41+00:00; user=hwnrwx; artist=Legião Urbana; excluded=False	True
1	hwnrwx | 1600151836 | Legião Urbana	UTC timestamp parses; user and artist text are trimmed; row remains a valid listen event.	timestamp=2020-09-15 06:37:16+00:00; user=hwnrwx; artist=Legião Urbana; excluded=False	True
2	hwnrwx | 1600151634 | Legião Urbana	UTC timestamp parses; user and artist text are trimmed; row remains a valid listen event.	timestamp=2020-09-15 06:33:54+00:00; user=hwnrwx; artist=Legião Urbana; excluded=False	True
49	hwnrwx | 1600088683 | Tyler, The Creator, Wanya Morris, Dâm-Funk, Austin Feinstein, Syd	UTC timestamp parses; user and artist text are trimmed; row remains a valid listen event.	timestamp=2020-09-14 13:04:43+00:00; user=hwnrwx; artist=Tyler, The Creator, Wanya Morris, Dâm-Funk, Austin Feinstein, Syd; excluded=False	True
48	hwnrwx | 1600088938 | Tyler, The Creator, Aaron Shaw, Samantha Nelson, Austin Feinstein	UTC timestamp parses; user and artist text are trimmed; row remains a valid listen event.	timestamp=2020-09-14 13:08:58+00:00; user=hwnrwx; artist=Tyler, The Creator, Aaron Shaw, Samantha Nelson, Austin Feinstein; excluded=False	True

Limitations and remaining issues

This 50-listen sample is one public user's history, concentrated in a short historical window. It does not represent mainstream Spotify listeners, does not identify an intervention, and does not prove that an artist is commercially emerging. A production analysis would require a broader population, an explicit emerging-artist definition, a time horizon suitable for growth comparisons, and validated audience-segment features. No table merge was performed, so merge-specific checks remain not applicable.

GitHub branch: kunjesh Branch URL: https://github.com/kunjeshgupta/dso576_project/tree/kunjesh Commit ID: fff1c088381a946bd1d2fae77593fc72bf441037 (implementation commit)